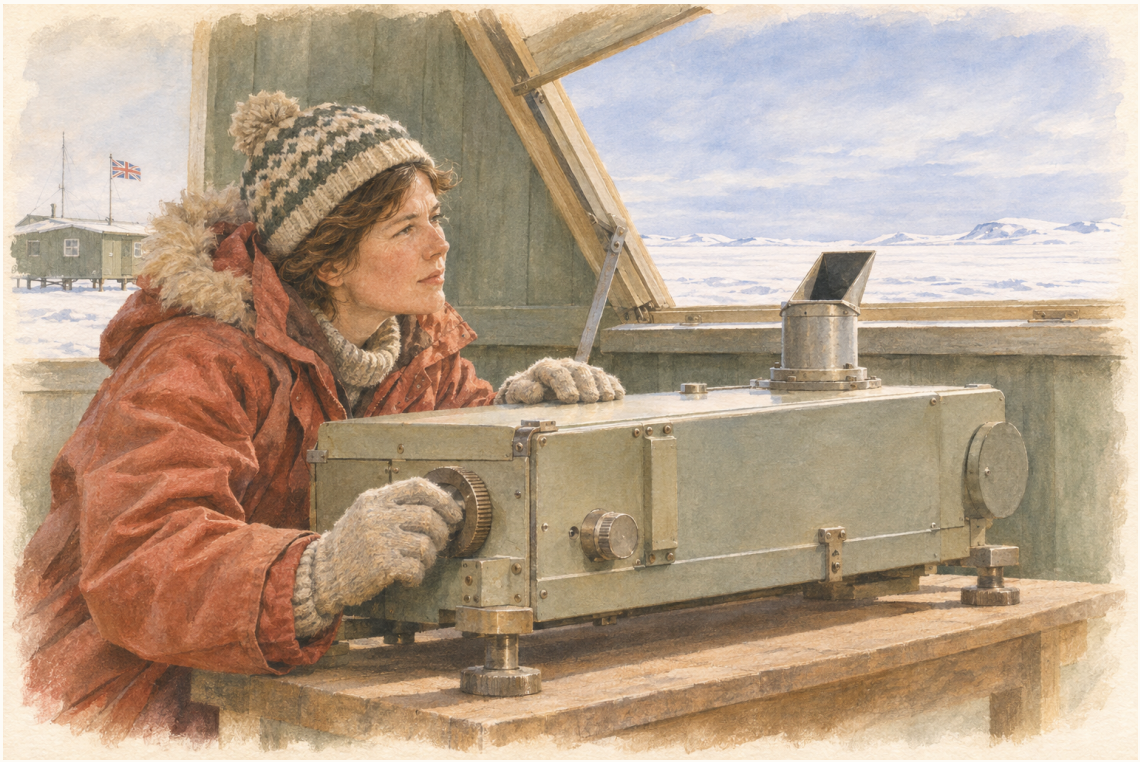
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拯救天空

Saving the Sky



臭氧层的低值，为什么被软件怀疑?

英文原文

A **layer** of **ozone shields** Earth **from** much of the sun's **ultraviolet radiation**. In June 1974 Mario Molina and Sherwood Rowland, chemists in California, warned in Nature of a **menace** in fridges and **spray** cans. Chlorofluorocarbons (CFCs) are so **inert** that they drift **intact** into the stratosphere, high in the **atmosphere**, where they can **liberate chlorine**, a **catalyst** that destroys ozone **molecule** after molecule. The pair foresaw **gradual erosion**.

The **alarm**, when it sounded, was **abrupt**. In May 1985 Joseph Farman and colleagues at the British **Antarctic** Survey reported that **ozone** over Halley Bay, their Antarctic station, had **plummeted** each southern spring since the mid-1970s. Their **humble** Dobson spectrophotometer **gauged** ozone from the ground. The staff were **persistent**, reading it in a **painstaking routine** of **manual observation**.

Yet NASA's Nimbus-7 **satellite**, a **sophisticated** eye **in orbit**, had recorded the same lows. Its software **flagged** them **as dubious**, since ozone that low had never been reliably measured. NASA says it was still **verifying** them when Farman published. The **sensor** had worked; **expectation** had not.

Remarkably, the response was **swift**. In September 1987 **diplomats** signed the Montreal **Protocol**, a **binding accord** to **curb** CFCs. **Amendments tightened** it: in June 1990 **affluent** countries **pledged** to **phase out** CFCs by 2000. In September 2009 the Protocol and its parent Vienna Convention became the UN's first **treaties** to win **universal ratification**.

In January 2023 a UN-backed **assessment forecast recovery** to 1980 levels around 2066 over **Antarctica**, 2045 over the **Arctic**, and 2040 elsewhere. **Healing** will take decades. The sky's **rescue** began with **diligent** observers who trusted that a **baffling** number was no **glitch**, and published it.

中文译文

一层臭氧为地球遮挡了太阳的大部分紫外辐射。1974年6月，马里奥·莫利纳和舍伍德·罗兰，两位化学家，在加州工作，在《自然》杂志上撰文警告，有一种威胁藏在冰箱和喷雾罐里。氯氟烃（CFCs）如此惰性，以至于它们能完好无损地飘入到平流层中，也就是高层大气中，在那里，它们可能释放出氯，氯是一种催化剂，会把臭氧分子一个接一个地破坏掉。两人预料到的，是一场渐进式的侵蚀。

警报真的拉响时，却来得十分突然。1985年5月，约瑟夫·法曼和英国南极调查局的同事报告称，哈雷湾上空的臭氧，那是他们的南极考察站，自20世纪70年代中期起，每到南半球春季就骤降。他们那台不起眼的多布森分光光度计，从地面测量臭氧。工作人员持之以恒，读数时遵循一套一丝不苟的例行规程，全靠人工观测。

然而，美国国家航空航天局（NASA）的雨云7号卫星，这只精密的眼睛在轨道上运行，也记录到了同样的低值。它的软件把这些读数标记为可疑，因为这么低的臭氧此前从没有被可靠地测到过。据NASA的说法，它还在核实这些读数，就在法曼发表论文的时候。传感器并没有出问题；预期却出了问题。

引人注目的是，各国的反应十分迅速。1987年9月，外交官们签署了《蒙特利尔议定书》，这是一项具有约束力的协定，用来限制CFCs。此后的修正案又收紧了它的管控：1990年6月，富裕国家承诺在2000年前逐步淘汰CFCs。2009年9月，该议定书与其所依据的《维也纳公约》一道，成为联合国首批获得普遍批准的条约。

2023年1月，一份由联合国支持的评估报告预测，臭氧层将恢复到1980年的水平：约2066年在南极上空，2045年在北极上空，其他地区则在2040年。臭氧层的愈合还需要数十年。天空得以拯救，始于一群勤勉的观测者，他们相信，一个令人费解的数字并不是小故障，于是把它发表了出来。

词汇与表达

layer n.
层

ozone n.
臭氧
layer of ozone shields Earth from much

shields from v.
为……遮挡了

ultraviolet adj.
紫外

radiation n.
辐射

menace n.
威胁
of a menace in fridges and spray cans.

spray n.
喷雾

inert adj.
惰性
inert that they drift intact into the

intact adj.
完好无损地
inert that they drift intact into the

atmosphere n.
大气

liberate v.
释放
where they can liberate chlorine,

chlorine n.
氯

catalyst n.
催化剂
catalyst that destroys ozone molecule

molecule n.
分子

gradual adj.
渐进式的

erosion n.
侵蚀
The pair foresaw gradual erosion.

alarm n.
警报

abrupt adj.
突然
alarm, when it sounded, was abrupt.

Antarctic adj.
南极

plummeted v.
骤降
plummeted each southern spring since

词汇与表达（续）

humble adj.
不起眼的

gauged v.
测量

persistent adj.
持之以恒
The staff were persistent, reading it

painstaking adj.
一丝不苟的
it in a painstaking routine of manual

routine n.
例行

manual adj.
人工

observation n.
观测

satellite n.
卫星

sophisticated adj.
精密的
a sophisticated eye in orbit,

in orbit n.
在轨道上运行

flagged as phr.
把……标记为

dubious adj.
可疑
Its software flagged them as dubious,

verifying v.
核实
was still verifying them when Farman

sensor n.
传感器

expectation n.
预期

Remarkably adv.
引人注目的是

swift adj.
迅速

diplomats n.
外交官

Protocol n.
议定书

binding adj.
约束力的
a binding accord to curb CFCs.

词汇与表达（续）

accord n.
协定
a binding accord to curb CFCs.

curb v.
限制
a binding accord to curb CFCs.

Amendments n.
修正案

tightened v.
收紧了

affluent adj.
富裕

pledged v.
承诺
June 1990 affluent countries pledged

phase out phr.
逐步淘汰
countries pledged to phase out CFCs by

treaties n.
条约

universal adj.
普遍

ratification n.
批准
to win universal ratification.

assessment n.
评估
2023 a UN-backed assessment forecast

forecast v.
预测

recovery n.
恢复

Antarctica n.
南极

Arctic n.
北极

Healing n.
愈合

rescue n.
拯救

diligent adj.
勤勉的

baffling adj.
令人费解的
a baffling number was no glitch, and

glitch n.
小故障

文章简述

- 预警：臭氧层能挡住大量紫外线。1974年，化学家警告：冰箱和喷罐中使用的氯氟烃会飘到高空，释放氯，持续破坏臭氧。
- 发现：1985年，南极地面观测人员报告，哈雷湾臭氧自70年代中期起，每逢南半球春季便骤降。
- 误判：卫星也测到同样低值，软件却因数值太罕见而标为可疑；NASA称当时尚在核实。仪器没有失灵，人的预期才出了错。
- 合作：各国1987年签署《蒙特利尔议定书》，限制氯氟烃；1990年富国承诺2000年前淘汰，议定书2009年获普遍批准。
- 恢复：2023年评估预测，臭氧恢复至1980年水平还需时间：其他地区约2040年、北极约2045年、南极约2066年。